VIT
Vellore Institute of Technology
CHENNAI

Reg. Number:	23BCE1463

Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	B. Tech	Semester	:	FALL 2025-26
Course Code & Course Title	:	BMAT205L & Discrete Mathematics and Graph Theory	Class Number	:	CH2025260100886, CH2025260100887, CH2025260100889, CH2025260100890, CH2025260100892, CH2025260100894, CH2025260100895, CH2025260100896, CH2025260100900, CH2025260100901, CH2025260100902
Faculty	:	Dr. Berin Greeni, Dr. Amit Kumar Rahul, Dr. Soumendu Roy, Dr. Sandip Dalui, Dr. Rajesh Kumar Mohapatra, Dr. Vidhya V, Dr. Hannah Grace G, Dr. Radha S, Dr. Pulak Konar, Dr. Kriti Arya, Dr. Ramesh Babu N	Slot	:	C2+TC2+TCC2
Duration	:	90 minutes	Max. Mark		50

General Instructions:

Write only your registration number on the question paper in the box provided and do not write other information

- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1.	(a)	Use predicate calculus to show that the premises "One student in this class knows how to write programs in JAVA" and "Everyone who knows how to write programs in JAVA can get a high-paying job" imply the conclusion "Someone in this class can get a high-paying job".	6	1	K3
	(b)	Using indirect method prove that $p \rightarrow q, \sim q \vee r, \sim r, \sim(\sim p \wedge s) \Rightarrow \sim s$.	4	1	K3
2		Without using the truth table obtain the PDNF and PCNF of $(a)\ p \wedge \sim(q \wedge r) \vee (p \rightarrow q)$ [5] $(b)(q \vee (p \vee r)) \wedge \sim((p \vee r) \vee q)$ [5].	10	1	K3
3	(a)	Let $(G,*)$ be an abelian group. Does the subset $H = \{a \in G : a = a^{-1}\}$ of G, where a^{-1} denote the inverse of the element a under the operation * form a subgroup of $(G,*)$ under the operation *? Justify your answer.	5	2	K3
	(b)	Find the code words generated by the encoding function $e: B^2 \rightarrow B^5$ where the parity check matrix is $H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$	5	2	K3
4	(a)	Consider the group $G_1 = \mathbb{R} \times \mathbb{R} = \{(a,b): a,b \in \mathbb{R}\}$ under the operation addition $'+'$ where $(a,b)+(c,d) = (a+c, b+d), \forall (a,b),(c,d) \in \mathbb{R} \times \mathbb{R}$. Let $G_2 = \mathbb{R}$ be the group under usual addition where $\mathbb{R}$ denote the set of all real numbers. Define $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ by	4	2	K4

		$f(x,y) = 2x - y$. Show that f is a group homormorphism.			
	(b)	Let G be the set of all elements that are coprime with 16. Prove that G forms a group under multiplication modulo 16. How many elements are self-inverses? Hint: Two elements a and b are said to be coprime if $gcd(a,b) = 1$.	6	2	K4
5	(a)	Find the number of permutations of the word "COMMITTEE" such that all vowels are together.	4	3	K2
	(b)	John owns six-coloured pairs of shoes (two red, two blue and two black). He wants to put all these pairs of shoes on the shoe rack. How many different arrangements of shoes are possible?	3	3	K2
	(c)	In a small village, there are 87 families, of which 52 families have at most 2 children. In a rural development programme 20 families are to be chosen for assistance, of which at least 18 families must have at most 2 children. In how many ways can the choice be made?	3	3	K2

**********All the best **********